

Midterm Project

Image Classification with Machine Learning Pipeline

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Project Overview

- **Classify digit in Extended MNIST Dataset**
 - <https://www.kaggle.com/datasets/crawford/emnist/data>
- Process raw dataset
- Implement HOG feature extraction
- Implement Support Vector Machine classifier
- Evaluate the performance matrices

EMNIST Dataset

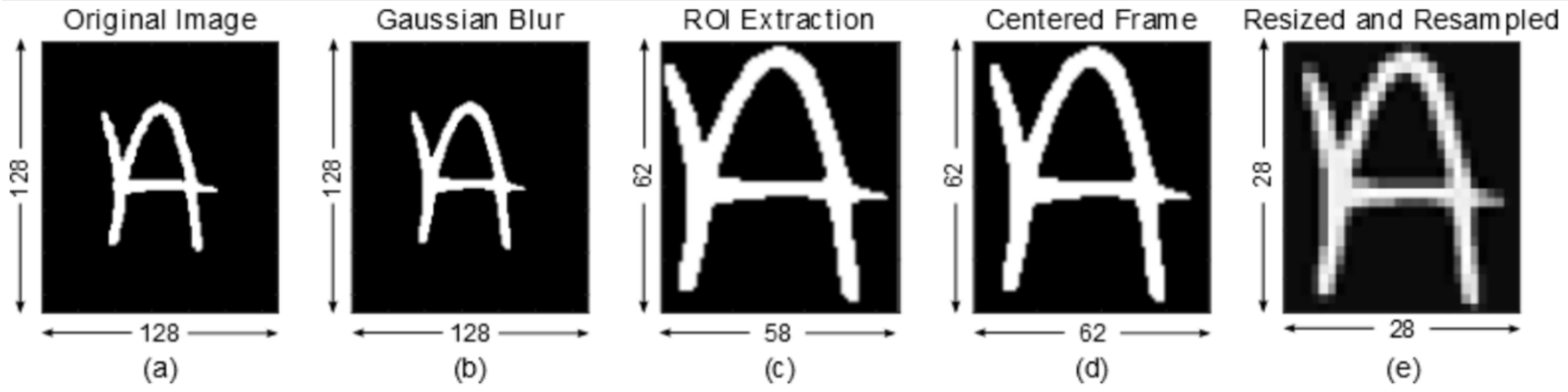


TABLE I
BREAKDOWN OF THE NUMBER OF AVAILABLE TRAINING AND TESTING
SAMPLES IN THE NIST SPECIAL DATABASE 19 USING THE ORIGINAL
TRAINING AND TESTING SPLITS.

	Type	No. Classes	Training	Testing	Total
By Class	Digits	10	344,307	58,646	402,953
	Uppercase	26	208,363	11,941	220,304
	Lowercase	26	178,998	12,000	190,998
	Total	62	731,668	82,587	814,255
By Merge	Digits	10	344,307	58,646	402,953
	Letters	37	387,361	23,941	411,302
	Total	47	731,668	82,587	814,255
MNIST [1]	Digits	10	60,000	10,000	70,000

Dataset Format

- Binary (8 bit format)
- CSV
 - Each row is separate image
 - 785 column
 - First column = class_label
 - Each column after represents one pixel value (784 total for a 28 x 28 image)
- More detail see : <https://www.kaggle.com/datasets/crawford/emnist/data>
- **Focus on Letters Datasets**
 - *The EMNIST Letters dataset merges a balanced set of the uppercase and lowercase letters into a single 26-class task.*

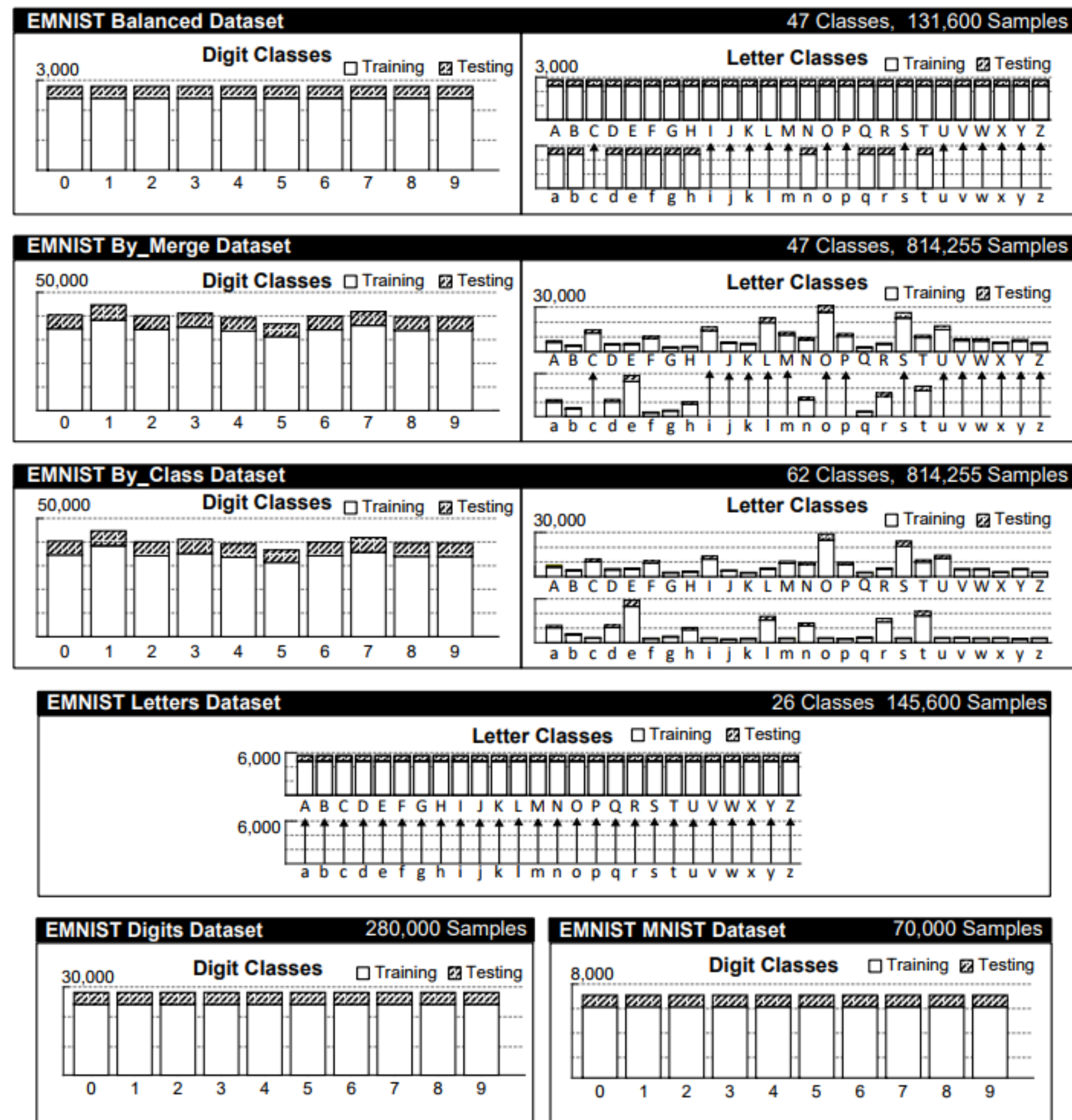


Fig. 2. **Visual breakdown of the EMNIST datasets.** The class breakdown, structure and splits of the various datasets in the EMNIST dataset are shown. Each dataset contains handwritten digits, handwritten letters or a combination of both. The number of samples in each class is shown for each dataset and highlights the large variation in the number of samples in the unbalanced datasets. The training and testing split for each class is also shown using either solid or hatched portions of the bar graphs. In the datasets that contain merged classes, a vertical arrow is used to denote the class into which the lowercase letter is merged.

1. Dataset Preparation

- Use 2600 samples only for this exam (make sure classes are ballance)
 - Each class has 100 samples
- Process dataset (binary or CSV) and visualize the image
- **Shuffle datasets before processing**
- Leave One Out Cross Validation (LOOCV) - Split Dataset
 - 80% for training
 - 20% for testing

2. Feature Extraction

- Use HOG Feature Extraction
- Change the default parameters
 - orientation
 - pixels_per_cell
 - cells_per_block

3. Classification

- Use Support Vector Machine (SVM) for classification
- Change default SVM parameters
 - Kernel type
 - Regularization (C)
 - Gamma
- Measure training performance
- **Use Grid Search Techniques to find the best performance**

4. Evaluation

- Evaluate on train (80%) and test datasets (20%)
- Performance metrics
 - accuracy
 - precision
 - recall
 - F1-score

Exam Submission

- Individual project
- Prefer to use Jupyter notebooks for doing this exam (run all before submit)
- Source code is submitted to your personal Github Account
- Make explanation video for source code and result
- Source code link and YT video submitted to learning
- **Due date : 18 May 2025**